



King Abdulaziz University

Faculty of Computing & Information Technology
Computer Science Department

Exam 01 (July 15th, 2018)
CPCS-223, Analysis and Design of Algorithm
Time: 09:30 am – 10:30am

Student ID: _____

Student Name: _____

***Section: _____

Instructor:

☐ Dr Umair

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Question 01

Which of the following is / are not the basic characteristic of an **Algorithm**?

- a) Step by Step Method
- b) Unambiguous Instruction
- c) Computational Procedure
- d) **Instances**

Question 02

Efficiency claim about an algorithm must be independent of

- a) Number of Instances
- b) **Bench Marks**
- c) Both a & b
- d) None

Question 03

The running time of “Selection Sort” is:

- a) **$O(n^2)$**
- b) $O(n)$
- c) $O(n \lg n)$
- d) $O(\lg n)$

Question 04

For RAM model, which one is not the basic assumption of the model?

- a) Infinitely large random-access memory
- b) Every instruction completes its execution in “unit” time
- c) **Instructions do not execute sequentially**
- d) Reproducible execution time

Question 05

What will be the value of $\sum_{k=0}^n m$

- a) $n * m$
- b) n
- c) $n + m$
- d) **$(n + 1) * m$**

Question 06

In “Worst Case” analysis, the running time of “Insertion Sort” is:

- a) **$O(n^2)$**
- b) $O(n)$
- c) $O(n \lg n)$
- d) $O(\lg n)$

Question 07

In “Best Case” analysis, the running time of “Insertion Sort” is:

- a) $O(n^2)$
- b) **$O(n)$**
- c) $O(n \lg n)$
- d) $O(\lg n)$

Question 08

Asymptotic notation describes the running time variations between “Upper” bound and “Lower” bound:

- a) **True**
- b) False

Question 09

Which of these is / are not “tight bounds”?

- a) **o**
- b) Θ
- c) Ω
- d) **ω**

Question 10

According to the definition, which condition(s) is / are correct for Theta-Notation?

- a) $f(n) > 0$**
- b) c_1 and c_2 must be positive integers**
- c) n can be negative
- d) It describes Upper Bound

Prove / disprove that $f(n) = \Omega g(n)$ for the given function

$$f(n) = an^2 + bn + c$$

Solution:

$$f(n) = an^2 + bn + c$$

and

$$g(n) = n^2$$

we have to prove

$$f(n) \geq d g(n)$$

$$an^2 + bn + c \geq d n^2$$

divide by “ n^2 ”

$$a + b/n + c/n^2 \geq d$$

It will hold for $d \geq a$ and $n_0 = 1$

Hence we can write

$$f(n) = \Omega g(n)$$

Write down the “loop invariant” of the following algorithm.

```
INSERTION-SORT(A)
1  for  $j \leftarrow 2$  to  $length[A]$ 
2      do  $key \leftarrow A[j]$ 
3          ▷ Insert  $A[j]$  into the sorted
              sequence  $A[1 \dots j - 1]$ .
4       $i \leftarrow j - 1$ 
5      while  $i > 0$  and  $A[i] > key$ 
6          do  $A[i + 1] \leftarrow A[i]$ 
7               $i \leftarrow i - 1$ 
8       $A[i + 1] \leftarrow key$ 
```

Solution:

“At the start of each iteration of the for loop the subarray $A[1 \dots j-1]$ consist of the elements originally in $A[1 \dots j-1]$ but in sorted order.”

Extra Sheet for Rough Work